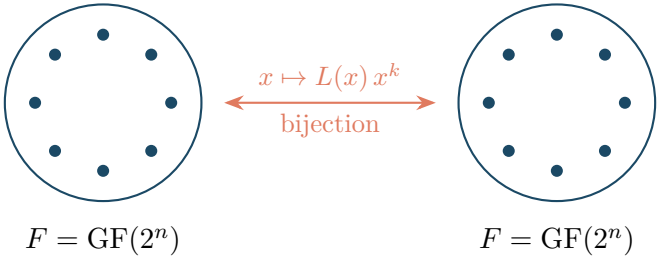
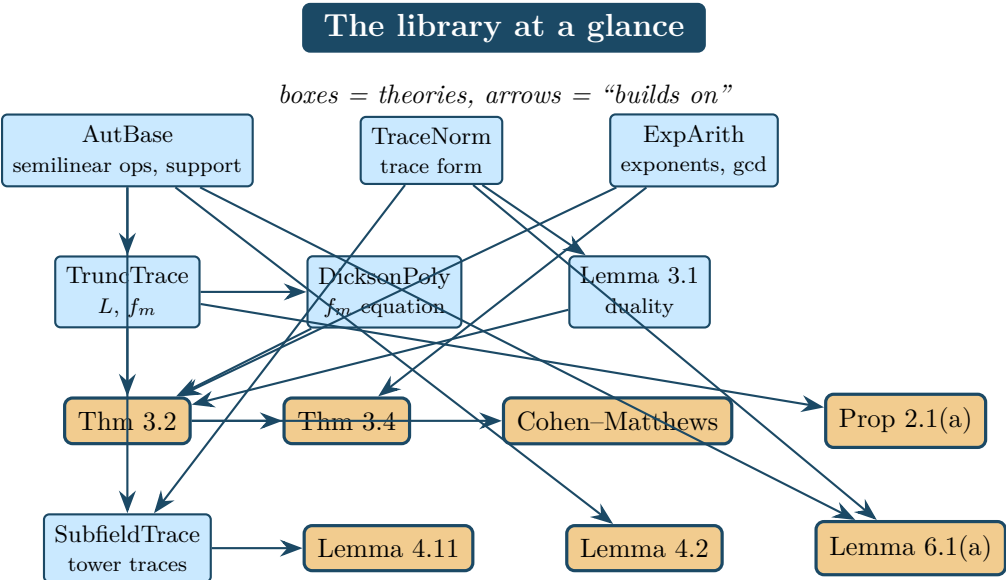


Pictures of the Patterns

Permutation polynomials & translation planes of even order
a visual atlas of everything the Lean library proves — few words, many diagrams

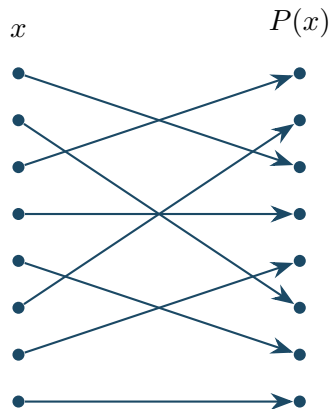


Everything below lives on one finite field of even order.
The recurring shape: a map that permutes the whole field.



1. The core object: a permutation polynomial

$$P(x) = L(x) \cdot x^k \text{ shuffles } F \text{ with no clashes and no gaps}$$



every input $\rightarrow$ a distinct output
(injective = surjective on a finite field)

$$P = \underbrace{L}_{\text{additive}} \times \underbrace{x^k}_{\text{power}} \quad \text{split}$$

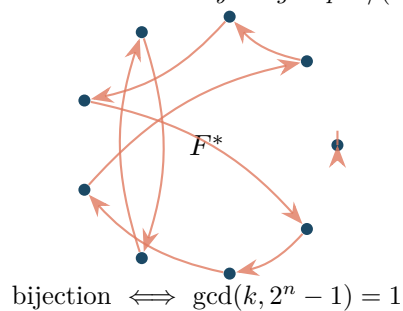
additive part L : a straight line through 0

$$L(a+b) = L(a) + L(b); \text{ if } P \text{ is a bijection then so is } L$$

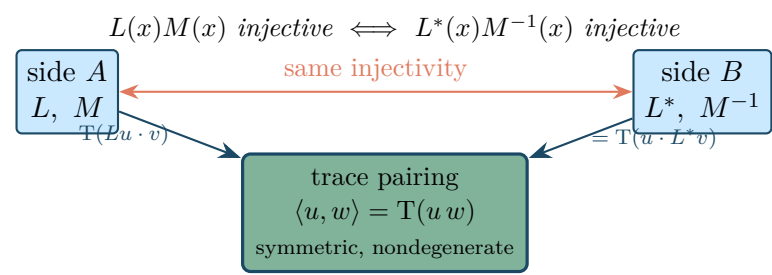
L keeps the additive grid; kernel = $\{0\} \Rightarrow$ bijection

power part x^k : winding the unit circle

on F^* it is $\times k$ on the cyclic group $\mathbb{Z}/(2^n - 1)$



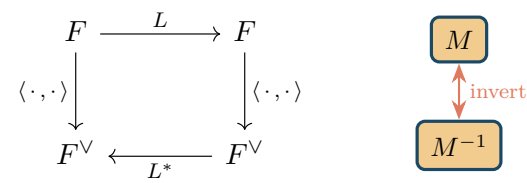
2. Lemma 3.1 — a duality (iso-equivalence)



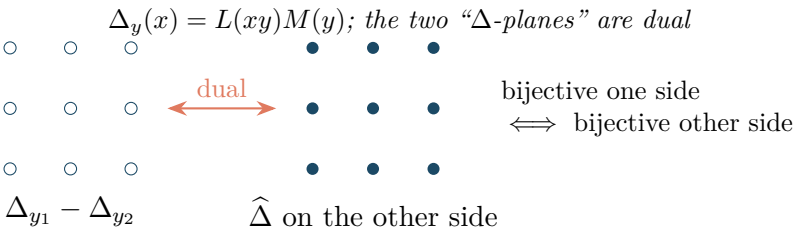
the adjoint square commutes

$$L \text{ bijective} \iff L^* \text{ bijective under the pairing}$$

mult. bijection

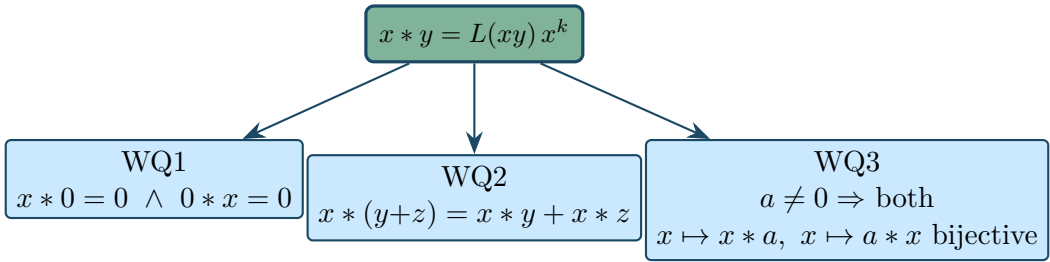


the pairing turns products into a shift



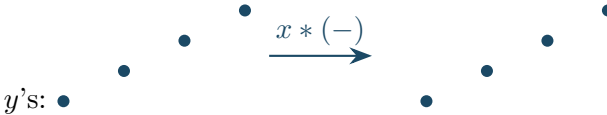
3. Prop 2.1(a) — a new multiplication

$x * y = L(xy) x^k$ makes F a weak quasifield



WQ2 = left line stays a line

multiplication by a fixed x is additive



colinear inputs $\rightarrow$ colinear outputs

WQ3 = Latin square

each nonzero row / column is a permutation

$a *$

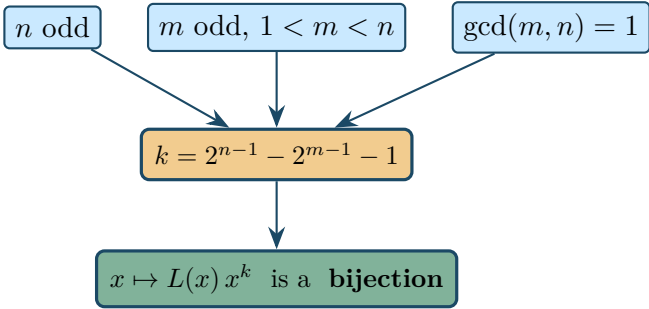
	0	1	2	3	4
	1	2	3	4	0
	2	3	4	0	1
	3	4	0	1	2
	4	0	1	2	3

$*b$

every symbol once
per row and column

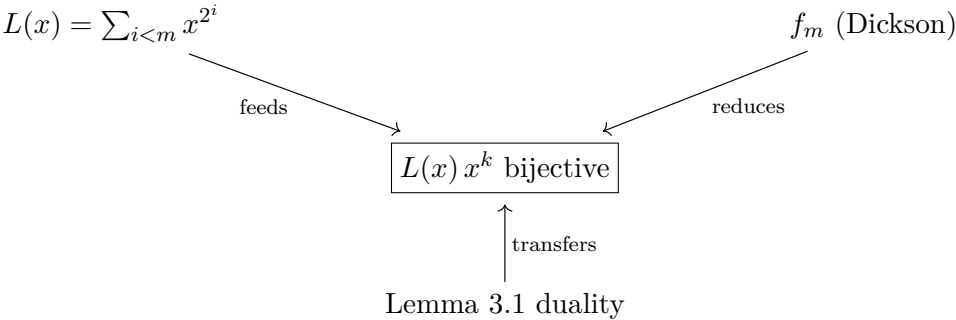
4. Theorem 3.2 — an explicit family

on $\text{GF}(2^n)$ with the right exponent, $L(x)x^k$ is a permutation



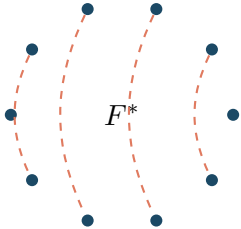
how it is built: three ingredients meet

truncated trace L , Dickson f_m , trace duality



Dickson functional equation

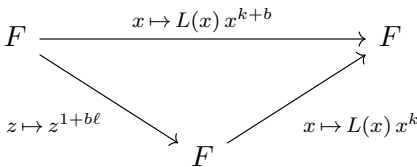
f_m separates y from y^{-1} on the units



f_m injective once $\{y, y^{-1}\}$ collapsed

5. Theorem 3.4 — functorial transfer

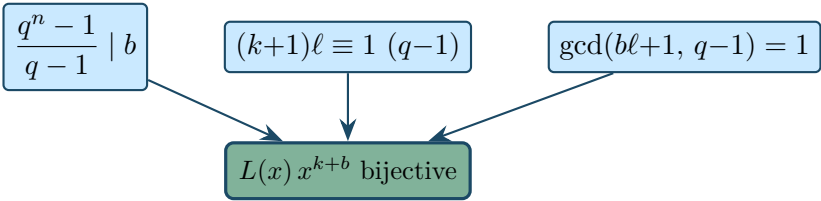
one permutation gives a whole family, via a commuting triangle



new map = (old permutation) $\circ$ (a power permutation) $\Rightarrow$ still a bijection.

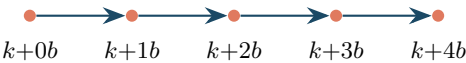
side conditions = gcd gates

the two factors are bijections exactly when their gcd's vanish



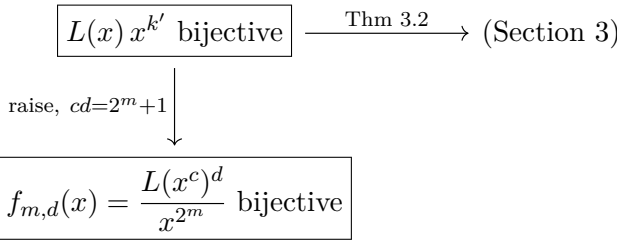
the family, pictured

shifting k by multiples of b keeps the bijection
each step: compose with a power map



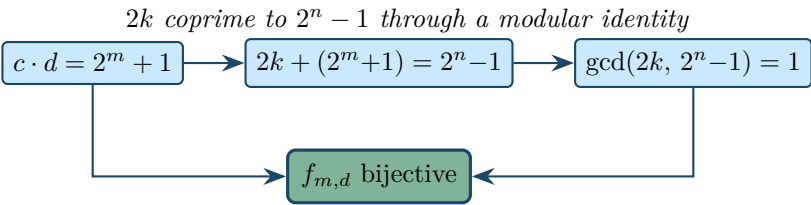
6. Cohen–Matthews connection

the same permutation reappears as a known bijection



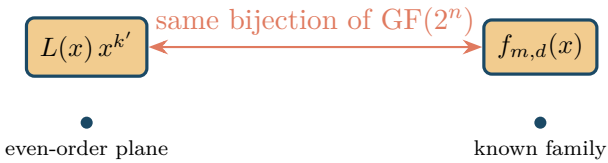
$cd = 2^m + 1$; the exponent bookkeeping (a gcd chain) is what makes it click.

exponent bookkeeping = a coprime chain



one map, two names

Dempwolff–Müller polynomial $\equiv$ Cohen–Matthews polynomial



7. Subfield traces — a tower

$T_{n:d}$ folds $\text{GF}(2^n)$ down onto $\text{GF}(2^d)$

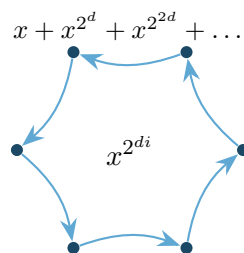


transitivity

$$T_{n:e} = T_{d:e} \circ T_{n:d}$$

the triangle commutes

$T_{n:d}$ = sum along a Frobenius orbit



add up one orbit $\rightarrow$ lands in $\text{GF}(2^d)$

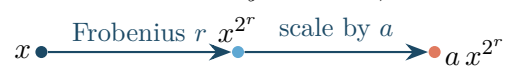
Lemma 4.11 — $\hat{T}$ commutes with scaling

$b \in \text{GF}(2^d)$ slides through the hatted trace

$$\begin{array}{ccc} x & \xrightarrow{\hat{T}_{n:e}} & \hat{T}_{n:e}x \\ b \cdot (-) \downarrow & & \downarrow b \cdot (-) \\ bx & \xrightarrow{\hat{T}_{n:d}} & b \cdot \hat{T}_{n:d}x \end{array}$$

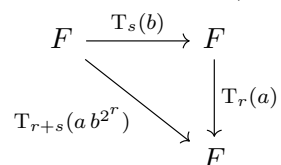
8. Semilinear operators $T_r(a)$

$x \mapsto a x^{2^r}$ — twist by Frobenius, then scale



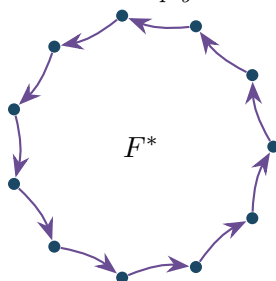
composition law

$$T_r(a) \circ T_s(b) = T_{r+s}(a b^{2^r})$$



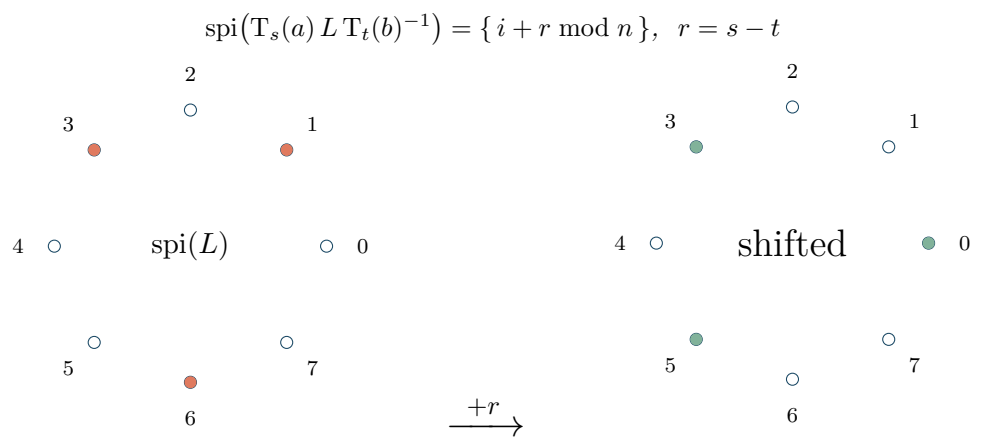
Singer group = one big cycle

$T_0(a) : x \mapsto ax$ acts sharply transitively on F^*



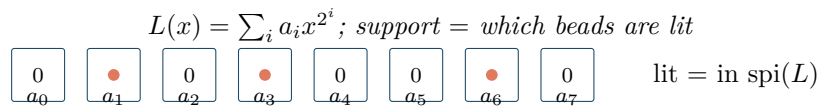
a generator visits every nonzero element once

9. Lemma 4.2 — conjugation shifts the support



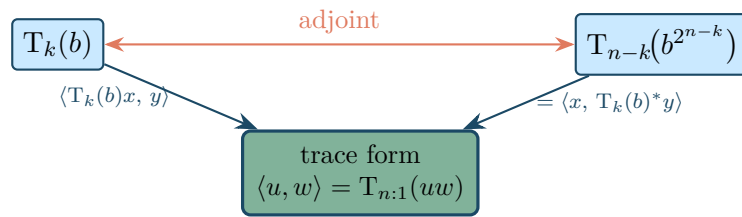
support beads rotate rigidly by r around $\mathbb{Z}/n$ — pattern preserved.

additive polynomial = beads on a necklace



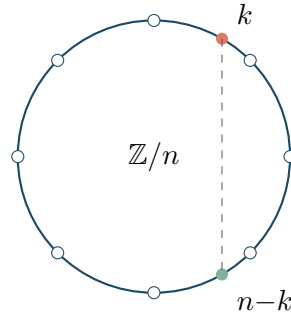
10. Lemma 6.1(a) — symplectic adjoint

under the trace form, $T_k(b)^* = T_{n-k}(b^{2^{n-k}})$



index reflection

$k \leftrightarrow n - k$ mirrored around the circle



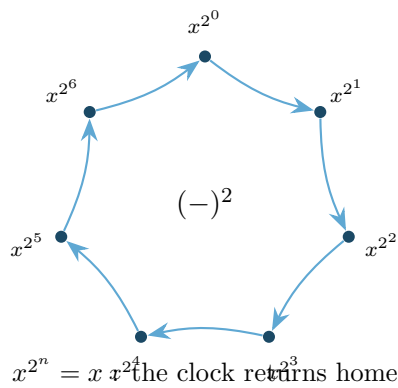
self-adjoint spread = symplectic

adjoint pairs the operators; the trace form is the glue

$$\begin{array}{ccc}
 F & \xrightarrow{T_k(b)} & F \\
 \langle \cdot, \cdot \rangle \downarrow & & \downarrow \langle \cdot, \cdot \rangle \\
 F^\vee & \xleftarrow{T_{n-k}(b^{2^{n-k}})} & F^\vee
 \end{array}$$

11. Frobenius — a clock of period n

$x \mapsto x^2$ cycles back after n steps on $\text{GF}(2^n)$



linearized polynomials ride the clock

Frobenius is additive: powers of 2 pass through sums

$$\begin{array}{ccc} (x+y) & \xrightarrow{(-)^{2^j}} & (x+y)^{2^j} \\ \parallel & & \parallel \\ x+y & \xrightarrow{(-)^{2^j}} & x^{2^j} + y^{2^j} \end{array}$$

Legend of pictures

how to read the shapes

• field element

○ empty slot / zero

→ a map

↔ an equivalence / duality

result

theory / hypothesis